TEXORA · LECTURE SHEET

Job, Contract and Process Costing

The three great costing methods worked numerically: a job cost sheet for a garment order, profit recognition on an incomplete contract, and a process account with normal loss, abnormal loss and scrap — each arithmetic chain shown in full.

AC-4205 · COST AND MANAGEMENT ACCOUNTING · TOPIC 4 · LEVEL 4 TERM 2

1. Job costing

In **job costing** each order is a separate cost unit: a job card collects its direct materials (from requisitions), direct labour (from time records) and direct expenses, then overheads are absorbed at the department rates of Topic 3, and administration and selling costs added to reach total cost. It suits production to a customer's specification — the garment export order is the textile archetype.

WORKED EXAMPLE — JOB COST SHEET, JOB 101 (2,000 JACKETS)

Direct material Tk 90,000 · Direct labour Tk 42,000 · Direct expenses Tk 3,000

Prime cost = 90,000 + 42,000 + 3,000 = **Tk 1,35,000**

Add factory overhead absorbed at 60 % of direct labour = 0.60 × 42,000 = Tk 25,200

Works cost = 1,35,000 + 25,200 = **Tk 1,60,200**

Add administration overhead at 10 % of works cost = Tk 16,020 → Cost of production = **Tk 1,76,220**

Add selling & distribution overhead at 5 % of cost of production = Tk 8,811 → Total cost = **Tk 1,85,031**

Quoted price with 20 % profit on cost = 1,85,031 × 1.20 = **Tk 2,22,037** (≈ Tk 111 per jacket).

The sheet's discipline: every rate is applied to its stated base, in sequence — prime, works, production, total, price.

2. Contract costing

Contract costing is job costing scaled up: one large, long-duration contract (a building, a bridge) is the cost unit, work proceeds at a site over several accounting periods, and the problem of the method is **how much profit to recognise on an incomplete contract**. Key quantities: **work certified** (valued by the contractee's architect/engineer), **work uncertified** (done but not yet certified, carried at cost), **retention money** (the percentage of certified value withheld until satisfactory completion), and

notional profit (certified value + uncertified cost − contract costs to date). Prudence governs recognition: no profit on barely-started contracts; on substantially advanced contracts a fraction of notional profit is credited, commonly:

$$\text{Profit taken} = \text{Notional profit} \times (2 \div 3) \times (\text{Cash received} \div \text{Work certified})$$

WORKED EXAMPLE — PROFIT ON AN INCOMPLETE CONTRACT

Contract price Tk 2,00,00,000. Work certified to date = 60 % of the price = **Tk 1,20,00,000**. The contractee pays 90 % of certified value in cash, retaining 10 %: cash received = **Tk 1,08,00,000** (retention Tk 12,00,000). Notional profit for the period = Tk 18,00,000.

Profit taken to the profit & loss account = 18,00,000 × (2/3) × (1,08,00,000 ÷ 1,20,00,000)
= 18,00,000 × 0.6667 × 0.90 = **Tk 10,80,000**.

The remaining Tk 7,20,000 of notional profit stays as a reserve in the work-in-progress account against future losses on the contract — profit is recognised as it is earned and collected, not as it is hoped.

3. Process costing

Where production flows continuously through sequential processes (dyeing → drying → finishing; or in chemical manufacture), cost is accumulated **per process per period** and averaged over output. The features that define the method: output of one process is the input of the next; **normal loss** (inherent shrinkage/evaporation, expected as a percentage of input) is borne by good output, its scrap value credited to the process; **abnormal loss** (beyond expectation) is costed at the full unit cost and charged out as a loss; **abnormal gain** is treated symmetrically.

$$\text{Cost per unit of good output} = (\text{Total process cost} - \text{Scrap value of normal loss}) \div (\text{Input units} - \text{Normal loss units})$$

WORKED EXAMPLE — A PROCESS ACCOUNT WITH ABNORMAL LOSS

Input 10,000 kg of material. Process costs: material Tk 5,00,000, labour Tk 1,20,000, overheads Tk 80,000 → total **Tk 7,00,000**.

Normal loss = 5 % of input = 500 kg, sold as scrap at Tk 10/kg → credit Tk 5,000.

Expected output = 10,000 − 500 = 9,500 kg.

Cost per kg = (7,00,000 − 5,000) ÷ 9,500 = 6,95,000 ÷ 9,500 = **Tk 73.16 per kg**.

Actual output = 9,200 kg → abnormal loss = 9,500 − 9,200 = 300 kg.

Value of abnormal loss = 300 × 73.16 = **Tk 21,947** (charged to the costing profit & loss account).

Value of output transferred on = 9,200 × 73.16 = **Tk 6,73,053**.

Check: 6,73,053 + 21,947 = 6,95,000 = net process cost ✓.

> The logic to remember: normal loss units carry no cost (their cost spreads over good units, raising unit cost to Tk 73.16), while abnormal loss units are costed exactly like good units — that is what makes the loss visible in money.

4. The three methods compared

FEATURE	JOB	CONTRACT	PROCESS
Cost unit	One order	One contract	One process for a period
Production	To customer specification, discontinuous	Large, long-term, at site	Continuous, standardised, for stock
Cost finding	Per job card, on completion	Per contract account, periodically	Average cost per unit per period
Special problems	Overhead absorption per job	Profit on incomplete contracts; retention	Normal/abnormal loss; joint products (Topic 5)
Textile examples	Export garment order	(Rare in textiles; plant installation jobs)	Spinning, dyeing, finishing lines

Glossary

Job card — the document accumulating one job's costs. **Work certified** — completed work valued by the contractee's representative. **Retention money** — the withheld percentage of certified value. **Notional profit** — profit shown by an incomplete contract account before prudence adjustments. **Normal loss** — inherent, expected process loss. **Abnormal loss/gain** — loss beyond / saving against the normal expectation, separately valued and reported.

Key points

• Job 101: prime Tk 1,35,000 → works Tk 1,60,200 → production Tk 1,76,220 → total Tk 1,85,031 → price Tk 2,22,037 at 20 % on cost.
• Incomplete contracts: profit taken = notional profit × 2/3 × cash/certified ratio = Tk 10,80,000 of Tk 18,00,000; the rest is reserved.
• Process costing: unit cost = (cost − scrap of normal loss) ÷ expected output = Tk 73.16/kg; abnormal loss is valued at that full unit cost (Tk 21,947) so inefficiency shows as money.

Practice

1. A process takes in 5,000 kg. Total process cost is Tk 3,00,000. Normal loss is 10 % of input, with scrap value Tk 20/kg. Actual output is 4,300 kg. Compute the cost per kg, the value of output, and the abnormal loss in units and value.
2. On a contract, notional profit is Tk 9,00,000, work certified Tk 80,00,000 and cash received Tk 60,00,000. Find the profit to be credited using the 2/3 formula, and the amount held in reserve.
3. A job shows: direct material Tk 60,000, direct labour Tk 30,000, direct expenses Tk 5,000; factory overhead 50 % of direct labour; administration 10 % of works cost. Build to the cost of production.

Answer key

1. Normal loss = 500 kg; scrap credit = 500 × 20 = Tk 10,000. Expected output = 4,500 kg. Cost per kg = (3,00,000 − 10,000) ÷ 4,500 = 2,90,000 ÷ 4,500 = **Tk 64.44/kg**. Output value = 4,300 × 64.44 = **Tk 2,77,111**. Abnormal loss = 4,500 − 4,300 = **200 kg**, value = 200 × 64.44 = **Tk 12,889**. (Check: 2,77,111 + 12,889 = 2,90,000 ✓.)
2. Profit taken = 9,00,000 × (2/3) × (60,00,000 ÷ 80,00,000) = 9,00,000 × 0.6667 × 0.75 = **Tk 4,50,000**. Reserve = 9,00,000 − 4,50,000 = **Tk 4,50,000**.
3. Prime cost = 60,000 + 30,000 + 5,000 = Tk 95,000. Factory overhead = 50 % × 30,000 = Tk 15,000 → works cost = Tk 1,10,000. Administration = 10 % × 1,10,000 = Tk 11,000 → **cost of production = Tk 1,21,000**.

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